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HUJI Applicative Research Report — Week 45, 2026

4 high-potential paper(s) · Weekly · Week 45, 2026 · Generated November 02, 2026

This week's digest highlights groundbreaking advancements across several key therapeutic areas. Noteworthy innovations include novel molecular glue degraders targeting previously undruggable cancer proteins, and the development of high-potency, non-opioid analgesics for chronic pain. Further promising research showcases a regenerative biologic for joint repair and a precision medicine approach to enhance cystic fibrosis treatment outcomes for specific patient populations.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Raphael I Benhamou** — Drug Discovery, Proteomics, Clinical
- **Avi Priel** — Drug Discovery, Neuroscience, Clinical
- **Amir Haze** — Medical Device, Clinical, Synthetic Biology, Proteomics
- **Batsheva Kerem** — Drug Discovery, Genomics, Clinical

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Molecular glue mediated degradation of cannabinoid receptor 1 for cancer therapy. 33/50

"Molecular Glues: A New Frontier for Degrading "Undruggable" Cancer Proteins."

Main researcher: Raphael I Benhamou · raphael.benhamou@mail.huji.ac.il.

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Published in npj drug discovery · 2026-10-26

ABOUT THIS RESEARCH

Researchers are developing a new class of drugs called molecular glues to target and degrade the cannabinoid receptor 1 (CB1) to treat cancer. This approach aims to eliminate specific proteins that promote tumor growth rather than just inhibiting their function.

COMMERCIAL ANGLE

Developing first-in-class targeted protein degraders (TPDs) for oncology targets that were previously considered undruggable.

COMMERCIALISATION METRICS

Novelty 8/10

The application of molecular glue technology to target the Cannabinoid Receptor 1 represents a highly innovative shift from traditional antagonism to targeted protein degradation. This approach offers a novel mechanism to overcome potential resistance and provides a specialized therapeutic modality for cancer treatment.

Commercial Potential 8/10

Molecular glues are a high-value, platform-enabling modality in drug discovery, and targeting the CB1 receptor offers a novel, potentially licensable approach to oncology. The specific mechanism provides a clear path toward a proprietary small-molecule therapeutic candidate.

Market Size 7/10

Targeting the cannabinoid receptor 1 (CB1) for cancer therapy addresses a significant oncology market, though the therapeutic application of CB1 modulation in cancer is a specialized niche compared to broader oncology targets.

Tech Readiness 3/10

The paper describes a fundamental proof-of-concept for a novel modality (molecular glue) targeting a specific receptor, which typically implies early-stage discovery and preclinical validation rather than lead optimization or clinical trials.

IP Strength 7/10

Molecular glues targeting specific protein-protein interactions are highly patentable and offer strong chemical defensibility, though the specific novelty of targeting CB1 for oncology depends on the unique scaffold and binding mechanism.

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2. Carbamate-based cannabinoids uncover a conserved lipid-sensing pocket in nociceptive TRPA1 and TRPV1 channels. 33/50

"Next-Generation Cannabinoid Analgesics Offer Potent, Non-Opioid Pain Relief."

Main researcher: Avi Priel · avi.priel@mail.huji.ac.il.

The Institute for Drug Research (IDR), Department of Pharmacology, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Israel. Electronic address: avi.priel@mail.huji.ac.il.

Published in Pharmacological research · 2026-06

ABOUT THIS RESEARCH

Researchers have engineered new cannabinoid-based molecules that more effectively target TRPA1 and TRPV1 ion channels by adding a specific carbamate functional group. These modified compounds show significantly higher potency in activating these pain-sensing channels compared to natural CBD or CBG.

COMMERCIAL ANGLE

The development of these high-potency carbamate derivatives provides a structural roadmap for designing a new class of non-opioid analgesics for chronic pain management.

COMMERCIALISATION METRICS

Novelty 7/10

The research demonstrates significant chemical innovation by using rational design to transform known scaffolds into potent dual agonists, revealing a previously unknown conserved lipid-sensing mechanism. While it offers high target-validation value and potential for new drug classes, it builds incrementally on established phytocannabinoid chemistry rather than introducing an entirely new paradigm.

Commercial Potential 7/10

The research demonstrates successful rational design of small-molecule derivatives with enhanced potency and selectivity, representing a clear path toward a novel non-opioid analgesic lead series. However, the transition from structural discovery to a proprietary drug candidate requires significant validation of pharmacokinetic profiles and safety to reach high-tier commercial readiness.

Market Size 9/10

The research targets TRPA1 and TRPV1, two of the most high-value pathways in the multi-billion dollar non-opioid analgesic market. By developing potent dual-agonists, this work addresses a massive unmet medical need for chronic and acute pain management with significant commercial scalability.

Tech Readiness 3/10

The research is currently at the basic discovery/proof-of-concept stage, utilizing in vitro heterologous expression systems to validate molecular binding and potency. It lacks in vivo animal models, pharmacokinetic data, or toxicity profiling required to advance toward preclinical development.

IP Strength 7/10

The research demonstrates successful rational ligand design to create novel, highly potent carbamate-functionalized derivatives with improved selectivity, providing a clear pathway for composition-of-matter patent claims. While the chemical scaffolds are derivative of known phytocannabinoids, the specific structural modifications and the discovery of a conserved binding pocket offer significant defensibility for new non-opioid analgesic candidates.

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3. Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model. 33/50

"Regenerative Protein Shows Promise for Healing Cartilage and Bone Injuries."

Main researcher: Amir Haze

Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.; Orthopedic Surgery Department, Hadassah-Hebrew University Medical Center, Jerusalem, Israel.; The Wohl Institute for Translational Medicine, Hadassah-Hebrew University Medical Center, Jerusalem, Israel.

Published in Journal of orthopaedic research : official publication of the Orthopaedic Research Society · 2026-04

ABOUT THIS RESEARCH

Researchers tested a recombinant human amelogenin protein product called Remelix® in a goat model to treat cartilage and bone injuries. The study showed that the protein significantly promotes the healing of both cartilage and subchondral bone without adverse effects.

COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

COMMERCIALISATION METRICS

Novelty 4/10

The research focuses on validating a known biological mechanism (amelogenin) in a larger animal model rather than introducing a new molecular entity or novel platform. While the transition from rat to goat is a critical translational step, it represents incremental refinement of existing prior art rather than a groundbreaking scientific innovation.

Commercial Potential 8/10

The transition from rodent models to a large animal (goat) model significantly de-risks the technology and demonstrates clear readiness for clinical translation. The mention of a specific, branded product (Remelix®) and a defined carrier (PX 407) suggests a high degree of developmental maturity and a clear path toward a regulated medical device or biologic.

Market Size 8/10

The target market for osteochondral repair is substantial, driven by a high prevalence of joint injuries and the significant clinical unmet need for regenerative solutions in an aging population. The transition from rat to large animal models suggests a high potential for clinical translation into a large, high-value orthopedic market.

Tech Readiness 6/10

The study has progressed from small animal models to a large animal (goat) model, which is a critical regulatory milestone for orthopedic biologics. However, it remains in the preclinical stage and has not yet entered human clinical trials.

IP Strength 7/10

The use of a branded recombinant protein (Remelix®) suggests established manufacturing processes and likely composition-of-matter or process patents. However, the reliance on a well-known protein like amelogenin may face challenges regarding non-obviousness compared to existing recombinant growth factor technologies.

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4. SPL84 and trikafta® exhibit comparable and additive effects in patient-derived HBE cells carrying the 3849+10kb C→T/F508del CFTR variants. 31/50

"Precision Combination Therapy Improves Outcomes for Underserved CF Patients."

Main researcher: Batsheva Kerem · batshevak@savion.huji.ac.il.

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Published in Journal of cystic fibrosis : official journal of the European Cystic Fibrosis Society · 2026-06

ABOUT THIS RESEARCH

Researchers found that combining a new inhaled antisense oligonucleotide (SPL84) with the existing drug Trikafta® provides additive benefits for patients with a specific cystic fibrosis mutation. This combination therapy effectively restores CFTR protein function in human lung cells, potentially offering better clinical outcomes than current standards of care alone.

COMMERCIAL ANGLE

Developing a combination therapy or a precision medicine regimen that pairs existing CFTR modulators with targeted antisense oligonucleotides to capture the underserved segment of patients with modest responses to current treatments.

COMMERCIALISATION METRICS

Novelty 6/10

The research is an incremental optimization study focused on combination therapy rather than a fundamental discovery of a new biological mechanism or platform. While the additive effect of an ASO with an existing standard of care is clinically relevant, it represents an evolutionary application of known splicing modulation and CFTR corrector technologies.

Commercial Potential 8/10

The study demonstrates a clear additive therapeutic strategy for a specific, high-value patient population, providing a strong de-risking signal for a combination product approach. The focus on an inhaled ASO targeting a known clinical gap in current standard-of-care (Trikafta) represents a highly viable and direct path to clinical application.

Market Size 4/10

The target market is limited to a highly specific genetic sub-population (3849+10kb variant) within the already established cystic fibrosis patient base. While clinically valuable for non-responders, the addressable demand is niche and constrained by the prevalence of this specific splicing mutation.

Tech Readiness 6/10

The asset has already completed a Phase 2a clinical trial, indicating significant de-risking; however, the current paper focuses on ex vivo validation in patient-derived cells rather than advancing the clinical trial phase.

IP Strength 7/10

The innovation utilizes a specific antisense oligonucleotide (ASO) sequence targeting a known splicing defect, which is highly patentable and provides a clear pathway for composition-of-matter protection. However, the defensibility is slightly tempered by the fact that it is a combination therapy with an existing standard-of-care, which may face competition from other splicing modulators.

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